

where the total number of parameters is $1 + m$ with $m \geq 1$. Alternatively, we can express w_{de} as

$$w_{de}(z_{n-1} < z \leq z_n) = w(z_{n-1}) + \frac{w(z_n) - w(z_{n-1})}{z_n - z_{n-1}}(z - z_{n-1}), \quad (1 \leq n \leq m) \quad (6)$$

Now the parameters become $w(z_n)$'s, which are values of w_{de} at the divided points and boundaries z_n ($0 \leq n \leq m$). In this case we have

$$F(z_{n-1} < z \leq z_n) = e^{3\{[w(z_{n-1}) - w(0)] + \frac{w(z_n) - w(z_{n-1})}{z_n - z_{n-1}}(z - z_{n-1})\}} \left(\frac{1 + z}{1 + z_{n-1}} \right)^{3 \frac{w(z_{n-1})(1 + z_n) - w(z_n)(1 + z_{n-1})}{z_n - z_{n-1}}} \\ \times (1 + z)^3 \prod_{i=1}^{n-1} \left(\frac{1 + z_i}{1 + z_{i-1}} \right)^{3 \frac{w(z_{i-1})(1 + z_i) - w(z_i)(1 + z_{i-1})}{z_i - z_{i-1}}}, \quad (1 \leq n \leq m) \quad (7)$$

where we have used $z_0 = 0$. For w_{de} in the last bin $z \in (z_m, \infty)$, we set it to be a constant w_L , and

$$F(z > z_m) = F(z_m) \left(\frac{1 + z}{1 + z_m} \right)^{3(1 + w_L)} \quad (8)$$

Now the formula for $H(z)$ is ready.

There is one more thing to be mentioned: once we have fitted our model with the data introduced in the next section, errors of $w(z_i)$ are correlated, i.e., the errors of $w(z_i)$ are dependent on each other. New parameters can be defined by transforming the covariance matrix of $w(z_i)$, so that errors of new parameters are decorrelated and do not entangle with each other. The new uncorrelated parameters are referred to as the principal components [10, 23], and they are directly related to their own locations (unlike the correlated case). So errors of the uncorrelated parameters are more interpretable and meaningful. For more discussions and implications of the uncorrelated parameters, we refer to the references [11, 14, 24]. In section IV, we will show both errors of correlated and uncorrelated parameters of w_{de} . The uncorrelated technique we adopt from [11] is as follows.

1. Get the covariance matrix

$$C = \langle WW^T \rangle - \langle W \rangle \langle W^T \rangle \quad (9)$$

where W is the vector of $w(z_i)$. The Fisher matrix F is defined by $F = C^{-1}$.

2. Diagonalize the Fisher matrix by an orthogonal matrix O

$$F = O^T \Lambda O, \quad (10)$$